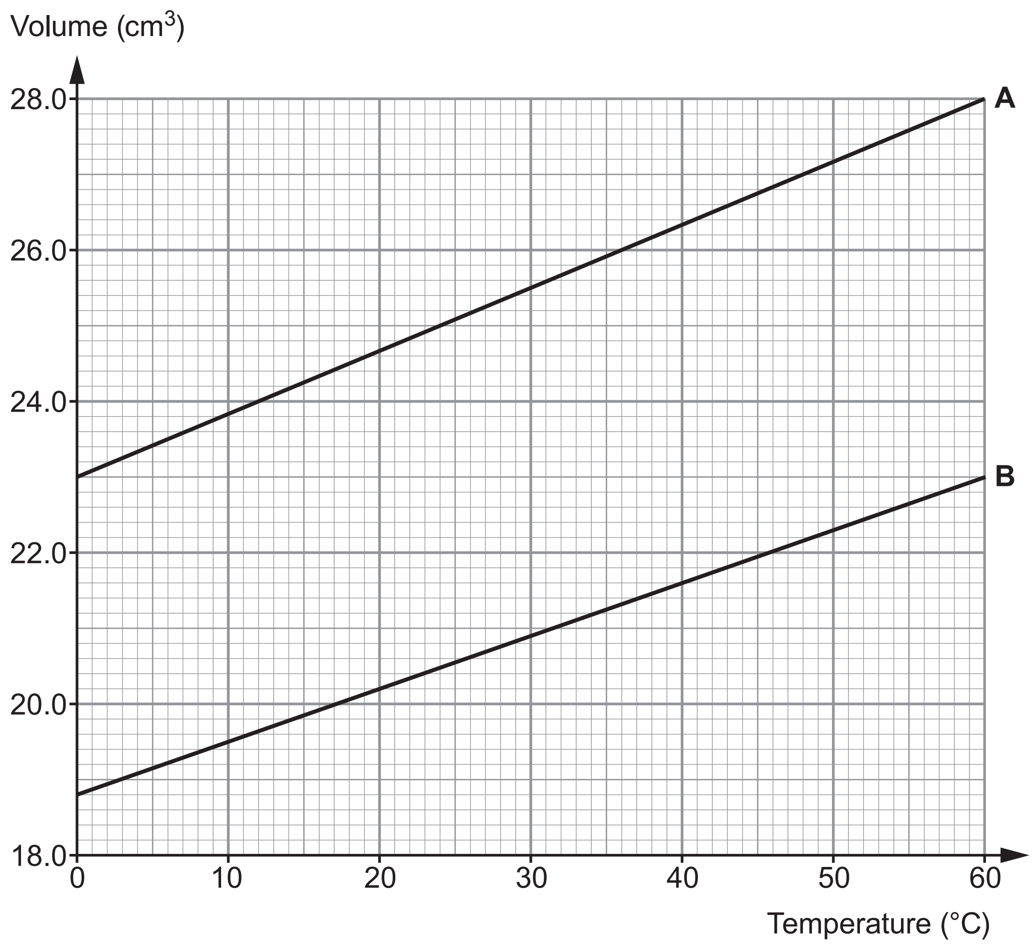
Examiner only

6. In a class experiment, the volumes of two different masses of trapped air (**A** and **B**) were measured as their temperature was increased from 0 °C to 60 °C. The results are shown in the graphs below.



(a) (i) At what temperature is the volume of gas **B** equal to 22.0 cm^3? [1]

Temperature = ... °C

(ii) The difference in the volumes of the two gases at 0 °C is 4.2 cm^3. Use the graph to calculate the difference in their volumes at 60 °C. [1]

Difference in volumes = ... cm^3

(iii) Both graph lines are straight. Using your answer to (ii) above, compare their gradients. [1]

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(b) (i) If both lines are extended to lower temperatures, they would meet at –273°C. This temperature is zero on the Kelvin scale of temperature. State the name given to this temperature. [1]

....

(ii) Calculate the highest temperature **on the Kelvin scale** reached by the gases in this experiment. [1]

Temperature = ... K

(c) In another experiment, gas **B** is kept in a container of fixed volume. State what happens, if anything, to the pressure of the gas as its temperature is increased. [1]

....

....

(d) (i) A gas exerts a force of 180 000 N on the walls of its container. The surface area of the container is 1.5 m^2. Use an equation from page 2 to calculate the pressure of the gas. [2]

Pressure = ... Pa

(ii) The gas is kept in a container at constant pressure. Julia says that if the area is doubled, the force would halve. Explain whether Julia's statement is correct. [2]

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